Numerical-Text Surrogate Modelling and Multi-Objective Optimisation of Compact Patch Antennas

Module 2 of the Applied Vision Transformer Framework for Hybrid-Slotted DGS Antennas in 5G and Emerging 6G Networks - Proof-of-Concept Prototype

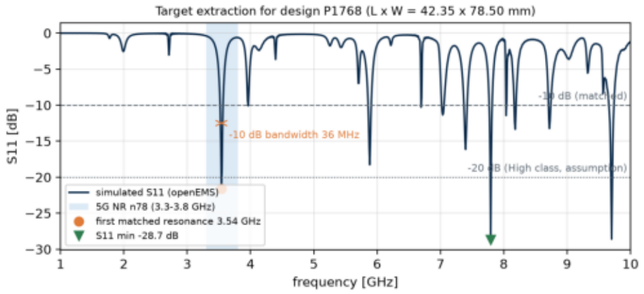
[Student name] | [Student ID] | [Programme / Module] | [Institution] | Applied AI Research Workshop - Module 2, October 2026

1 Research objective

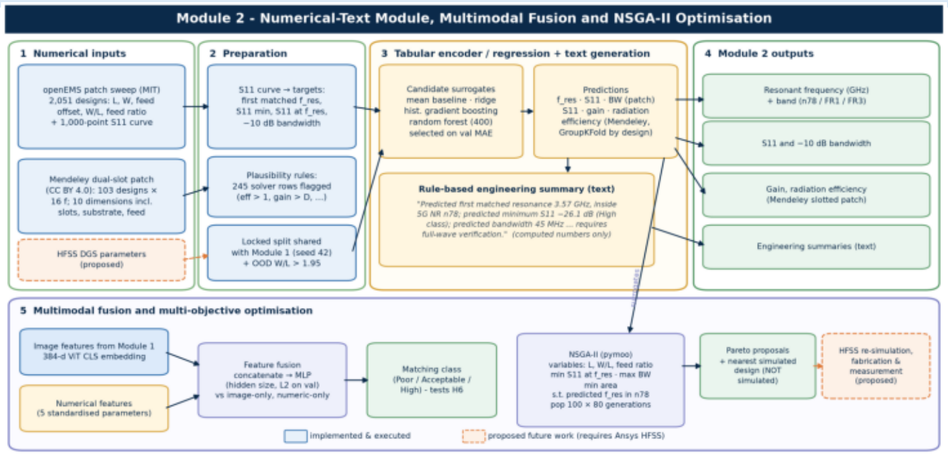
- How well do antenna parameters predict simulated resonance, S11, bandwidth, gain and efficiency?
- Do numbers add to Module 1's ViT image features? Can NSGA-II on surrogates propose n78 designs?
- Text summaries must never invent values.
- Proof of concept on public simulation data; Ansys HFSS was not available and was not run.

2 Data & preparation

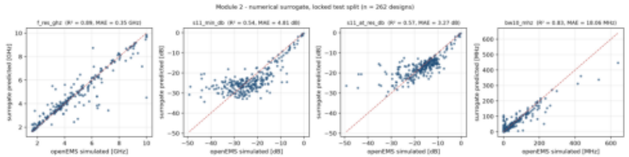
- openEMS patch sweep (Hugging Face, MIT): 2,051 designs; S11 curve -> 4 engineering targets
- Mendeley dual-slot patch v2 (CC BY 4.0): 103 x 16 f; 245 implausible solver rows flagged by a fixed rule
- Split shared with Module 1: 1,220 / 262 / 262 + OOD W/L > 1.95 (n = 307); GroupKFold for Mendeley
- Model selection on validation only; test scored once



3 Module 2 block diagram (implemented vs proposed)



4 Surrogate results (executed)



Target (random forest)	Test R²	MAE	OOD R²
First matched resonance [GHz]	0.89	0.35	0.44
Minimum S11 [dB]	0.54	4.81	0.13
S11 at resonance [dB]	0.57	3.27	-1.54
−10 dB bandwidth [MHz]	0.83	18.06	-3.72
Mendeley S11 [dB]	0.86	0.53	-
Mendeley Gain [dBi]	0.99	0.51	-
Mendeley Radiation efficiency [ratio]	0.97	0.03	-

Mendeley: out-of-fold, 5-fold GroupKFold by design; gain and efficiency on plausible rows only

Takeaways

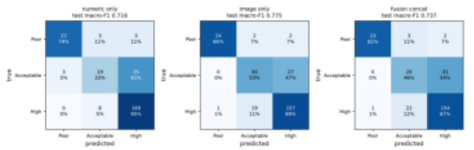
Resonance and bandwidth: well predicted

Matching depth: moderate (sharp impedance effect)

Wide OOD patches: fails -> search kept in range

Slot geometry halves the frequency-only error

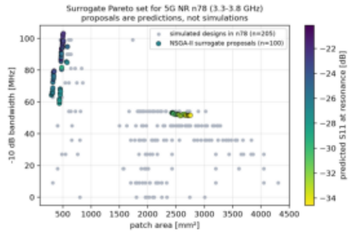
5 Multimodal fusion (H6)



Test macro-F1: fusion 0.737 | image-only 0.775 | numeric-only 0.716

H6 not supported. ViT embeddings alone beat Module 1's own head (0.581)

6 NSGA-II for 5G NR n78 (3.3-3.8 GHz)



100 Pareto proposals

area 299-2754 mm²

pred. BW 51-103 MHz

median 0.08 SD from a simulated design

Predictions only - NOT simulated

7 Outcomes & conclusion

E1	Surrogate beats mean baseline (all targets)	**Met**
E2	Slot geometry adds to frequency	**Met**
H6	Fusion beats image-only (pre-registered)	**Not supported**
E3	Feasible n78 proposals in range	**Met**

- Surrogates are only trustworthy where they were trained
- Next: space-filling HFSS DGS sweep + re-simulation

Fast, traceable surrogate screening of compact patch antennas for 5G / 6G - simulation evidence only, proposals not HFSS-validated

Data: LaBash et al. (2025) arXiv:2505.18188 [MIT] | Mihad, Islam & Sojib (2024) Antenna Parameters Dataset v2, doi:10.17632/3gxr2vvd9n.2 [CC BY 4.0]
Random forest: Breiman (2001) | NSGA-II: Deb et al. (2002) | pymoo: Blank & Deb (2020) | Code: https://github.com/rathiabeth/module2-numerical-surrogate-antenna-design